

HCSM-02

The Hodge Complex on Λ Differential Forms, the Hodge–Dirac Operator,
and the $\mathbb{Z}_2$ -Graded Structure of the Substrate

The Hodge Complex Standard Model

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September 2026

Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from the periodic 8×8 torus. This paper establishes the algebraic substrate on which the framework is built: the Hodge complex of differential forms on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. We construct the three graded pieces $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ of total dimension 256, define the exterior derivative d and its adjoint d^* , and prove that the Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree. Unlike the spinorial Dirac operator on the same lattice, the Hodge–Dirac square is the scalar 5–point Laplacian with no fermion doubling. The complex carries a canonical $\mathbb{Z}_2$ grading: even forms ($\Omega^0 \oplus \Omega^2$) are bosonic, odd forms (Ω^1) are fermionic, and D is the supercharge of an $\mathcal{N} = 1$ supersymmetric quantum mechanics. We prove the 14–fold lift: the scalar Laplacian’s multiplicity-14 eigenvalue at $\lambda = 4$ lifts to 14 modes in Ω^0 , 28 in Ω^1 , and 14 in Ω^2 , giving a 56–dimensional sector. By discrete Hodge theory, $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$. The full D_4 decomposition of the $\lambda = 4$ sector is computed: $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ with $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$. All theorems are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The Hodge complex in HCSM

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation.

The algebraic arena on which this identification is made is the *Hodge complex* of differential forms on Λ . This paper establishes the complex, its exterior derivative and codifferential, the Hodge–Dirac operator, and the $\mathbb{Z}_2$ -graded structure that will be used throughout the HCSM paper stack. The substrate is not a lattice of Majorana fermions with one fermion per site; it is the Hodge complex of differential forms, and the fermionic content lives in the odd form degree.

1.2 The one-line summary

The substrate of HCSM is the Hodge complex of differential forms on the 8×8 torus. Its Hodge–Dirac operator squares exactly to the scalar Laplacian, its $\mathbb{Z}_2$ grading pairs bosons and fermions at every eigenvalue, and its kernel has dimension 4 — the four directions of emergent spacetime.

1.3 The three central theorems

The paper is organized around three central theorems.

1. **Hodge–Dirac Square** (Section 3). On the discrete torus, the Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree $k = 0, 1, 2$, where Δ is the scalar 5–point Laplacian.
2. **Fourteen Fold Lift** (Section 5). The multiplicity-14 eigenvalue $\lambda = 4$ of the scalar Laplacian lifts to the full Hodge complex as $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$ modes.
3. **Kernel Dimension** (Section 6). By discrete Hodge theory, $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$. The four harmonic forms are the four emergent spacetime directions (see HCSM-03).

1.4 Cross references to the HCSM paper stack

The paper is foundational. It is cited by every other HCSM paper for the substrate algebra. Direct companions in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-00** [1]: foundations, notation, and mapping.
- **HCSM-01** [2]: derivation of the 8×8 torus substrate.
- **HCSM-03** [3]: Lorentzian signature from the Hodge star on $\ker D$.
- **HCSM-04** [4]: the fourteen-mode multiplet and its D_4 orbit decomposition.
- **HCSM-06** [5]: sector-coarse correspondence from the D_4 -invariant endomorphism algebra.
- **HCSM-12** [6]: the Hodge-HCSM interface, entropic flatness of the $\lambda = 4$ sector.

1.5 Organization

Section 2 states the substrate and the three form degrees. Section 3 constructs the Hodge–Dirac operator and proves $D^2 = \Delta$. Section 4 establishes the $\mathbb{Z}_2$ grading, its statistics, and the supercharge structure. Section 5 derives the spectrum and the 14-fold lift. Section 6 proves the kernel dimension. Section 7 computes the D_4 decomposition of the $\lambda = 4$ sector. Section 8 gives reproducibility notes. Section 9 concludes.

2 Setup: the substrate and the form degrees

2.1 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \tag{1}$$

consisting of $N = 64$ sites arranged in an 8-by-8 grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + 8, x_2)$ and with $(x_1, x_2 + 8)$; equivalently, the coordinates x_1 and x_2 are each defined modulo 8. Topologically, Λ is the torus $T^2 = S^1 \times S^1$.

Each site has four nearest neighbors: $(x_1 \pm 1, x_2)$ and $(x_1, x_2 \pm 1)$, with all coordinates taken modulo 8. The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e), \quad (2)$$

where the sum runs over the four unit lattice vectors $e = (\pm 1, 0), (0, \pm 1)$.

2.2 The three form degrees

On the two-dimensional torus Λ , there are three types of differential forms:

- Ω^0 : 0-forms (scalar functions on the 64 sites), dimension $\dim \Omega^0 = 64$, bosonic.
- Ω^1 : 1-forms (fields on the 128 directed edges, two components per site), dimension $\dim \Omega^1 = 128$, fermionic.
- Ω^2 : 2-forms (fields on the 64 plaquettes), dimension $\dim \Omega^2 = 64$, bosonic.

The total dimension is

$$\dim(\Omega^0 \oplus \Omega^1 \oplus \Omega^2) = 64 + 128 + 64 = 256. \quad (3)$$

We denote by Ω^k the k -form degree, $k = 0, 1, 2$, and by $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ the full Hodge complex.

2.3 The exterior derivative and the codifferential

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (4)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (5)$$

The codifferential $d^* = d^\dagger$ is the adjoint with respect to the standard Euclidean inner products on $\Omega^0, \Omega^1, \Omega^2$. Explicitly,

$$(d_0^* \omega)(x) = \sum_{\mu} [\omega_\mu(x) - \omega_\mu(x - e_\mu)], \quad (6)$$

$$(d_1^* \chi)(x, \mu) = \pm [\chi(x) - \chi(x + e_\mu)], \quad (7)$$

with the sign chosen so that $d_0^* d_1^* = 0$.

The exterior derivative is nilpotent:

$$d_1 d_0 = 0 \quad \text{exactly}, \quad (8)$$

and the codifferential satisfies $d_0^* d_1^* = 0$. This nilpotency is the algebraic foundation of the whole paper.

3 The Hodge–Dirac operator

3.1 Definition

Definition 3.1 (Hodge–Dirac operator). The Hodge–Dirac operator is

$$D = d + d^*, \quad (9)$$

acting on the full Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

In block form,

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (10)$$

a symmetric 256×256 real operator.

3.2 The Hodge–Dirac square

Theorem 3.2 (Hodge–Dirac square). *On the discrete torus Λ , the identity*

$$D^2|_{\Omega^k} = \Delta \quad (11)$$

holds exactly on each form degree $k = 0, 1, 2$, where Δ is the scalar 5-point Laplacian of (2).

Proof. Direct computation. On Ω^0 , D^2 reduces to $d_0^*d_0$. Expanding in components,

$$(d_0^*d_0f)(x) = \sum_{\mu} [f(x) - f(x - e_{\mu})] - \sum_{\mu} [f(x + e_{\mu}) - f(x)] = 4f(x) - \sum_{|e|=1} f(x + e), \quad (12)$$

which is Δf exactly. The same computation on Ω^1 gives $d_0d_0^* + d_1^*d_1 = \Delta \cdot \mathbb{K}_{\Omega^1}$, using the nilpotency $d_1d_0 = 0$ to remove cross terms. On Ω^2 , the identity reduces to $d_1d_1^* = \Delta$. All three identities are verified at machine precision in the companion repository. $\square$

Remark 3.3 (No fermion doubling). Unlike the spinorial Dirac operator on the same lattice, which satisfies $D_{\text{spin}}^2 = (\text{doubled Laplacian})$ and suffers from fermion doubling, the Hodge–Dirac square equals the scalar Laplacian exactly. This is the structural reason HCSM adopts the Hodge complex as its substrate: the Hodge–Dirac operator evades the doubling problem by construction. The lattice proves the Nielsen–Ninomiya evasion directly at the level of the differential-form complex.

4 The $\mathbb{Z}_2$ grading and the supercharge structure

4.1 The parity operator

Definition 4.1 (Parity operator). Define the parity operator $(-1)^F$ on the Hodge complex by

$$(-1)^F = \begin{cases} +1 & \text{on } \Omega^0 \oplus \Omega^2, \\ -1 & \text{on } \Omega^1. \end{cases} \quad (13)$$

The parity operator $\mathbb{Z}_2$ -grades the complex into a bosonic sector $\Omega^0 \oplus \Omega^2$ (even forms, eigenvalue $+1$) and a fermionic sector Ω^1 (odd forms, eigenvalue -1).

Proposition 4.2 (D is odd). *The Hodge–Dirac operator anticommutes with the parity operator:*

$$D(-1)^F = -(-1)^F D. \quad (14)$$

Proof. D maps Ω^k to $\Omega^{k\pm 1}$, changing the form degree by ± 1 and hence flipping the parity eigenvalue. Direct inspection of the block form (10) confirms the anticommutation. $\square$

4.2 Field quantization and statistics

Assign canonical commutation relations to even-sector modes a_n and canonical anticommutation relations to odd-sector modes ψ_n :

$$[a_n, a_m^\dagger] = \delta_{nm}, \quad \{\psi_n, \psi_m^\dagger\} = \delta_{nm}. \quad (15)$$

The Hodge–Dirac operator becomes the supercharge

$$D = \sum_n \sqrt{\lambda_n} \left(a_n^\dagger \psi_n + \psi_n^\dagger a_n \right), \quad (16)$$

where λ_n are the eigenvalues of Δ .

Theorem 4.3 (Form-degree statistics). *The three sectors of the Hodge complex carry the following statistics under field quantization:*

- Ω^0 (0-forms): bosonic;
- Ω^1 (1-forms): fermionic;
- Ω^2 (2-forms): bosonic.

Proof. The $\mathbb{Z}_2$ grading $(-1)^F$ defines the bosonic (even) and fermionic (odd) sectors. Quantization with commutators on the even sector and anticommutators on the odd sector gives the stated statistics. Fermionic statistics has a geometric origin: it is the $\mathbb{Z}_2$ -odd part of the Hodge complex. $\square$

4.3 The $\mathcal{N} = 1$ supersymmetric quantum mechanics algebra

Theorem 4.4 (Supercharge structure). *With $Q = D$ and $H = D^2/2 = \Delta/2$, the operators satisfy*

$$\{Q, Q^\dagger\} = 2H, \quad Q^2 = (Q^\dagger)^2 = 0. \quad (17)$$

This is the standard $\mathcal{N} = 1$ supersymmetric quantum mechanics algebra.

Proof. $Q^2 = (d + d^*)^2 = d^2 + \{d, d^*\} + (d^*)^2$. Since $d^2 = 0$ and $(d^*)^2 = 0$ by nilpotency, this reduces to $Q^2 = \{d, d^*\}$. But $\{d, d^*\}$ on each form degree is the Laplacian Δ , so $Q^2 = \Delta \cdot \mathbb{K} = 2H$. The anticommutator $\{Q, Q^\dagger\} = 2Q^2 = 4H$ differs only by the conventional normalization; with $H = \Delta/2$ the standard form (17) follows. $\square$

Corollary 4.5 (Boson–fermion pairing at every eigenvalue). *Every bosonic mode at energy λ_n has a fermionic partner at the same energy. In particular, at $\lambda = 4$:*

$$28 \text{ bosons} + 28 \text{ fermions at } \lambda = 4. \quad (18)$$

Proof. $\mathcal{N} = 1$ supersymmetric quantum mechanics pairs every bosonic eigenstate of H at energy E with a fermionic eigenstate at the same energy, unless the eigenstate is annihilated by Q or $Q^\dagger$. The $\lambda = 4$ sector is not in the kernel of Q (it is a nonzero eigenvalue of D^2), so the pairing is exact. $\square$

5 Spectrum and the 14-fold lift

5.1 The scalar Laplacian spectrum

The scalar Laplacian (2) on the 8×8 torus has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (19)$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues. The self-paired eigenvalue $\lambda = 4$ has multiplicity 14, corresponding to the 14 modes

$$\begin{aligned} &\{(0, 4), (4, 0)\}, \\ &\{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ &\{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (20)$$

These decompose under the D_4 point group of the square into three orbits of sizes $(2, 8, 4)$ (see HCSM-04).

5.2 The lift to the Hodge complex

Theorem 5.1 (Fourteen-fold lift). *The 14-fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as*

$$14 \text{ modes in } \Omega^0, \quad 28 \text{ modes in } \Omega^1, \quad 14 \text{ modes in } \Omega^2. \quad (21)$$

The total dimension of the $\lambda = 4$ sector is

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (22)$$

Proof. Since $D^2|_{\Omega^k} = \Delta$ by Theorem 3.2, the spectrum of D^2 on Ω^k is the scalar Laplacian spectrum with the indicated multiplicities. The Laplacian has multiplicity 14 at $\lambda = 4$ on Ω^0 ; on Ω^1 , each site has two 1-form components, so the multiplicity doubles to 28; on Ω^2 , the multiplicity is again 14. Summing gives 56. $\square$

Remark 5.2 (Uniqueness of $L = 8$). On an $L \times L$ torus, the multiplicity at the self-paired eigenvalue is exactly $2(L - 1)$. Setting this equal to 14 gives $L = 8$. The torus size is therefore selected by the requirement that one Standard Model generation sits at the self-paired eigenvalue, not chosen by hand.

6 The kernel of the Hodge–Dirac operator

6.1 Discrete Hodge theory

Theorem 6.1 (Kernel dimension). *The kernel of the Hodge–Dirac operator is isomorphic to the de Rham cohomology of the torus:*

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}), \quad (23)$$

with dimension

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \quad (24)$$

where (b_0, b_1, b_2) are the Betti numbers of the torus.

Proof. By the discrete Hodge theorem (Dodziuk 1976 [7]), the kernel of $D = d + d^*$ is isomorphic to the direct sum of the de Rham cohomology groups $H^k(T^2; \mathbb{R})$ for $k = 0, 1, 2$. The Betti numbers of the torus are $b_0 = 1$, $b_1 = 2$, $b_2 = 1$, so $\dim \ker D = 4$. $\square$

Remark 6.2 (The four harmonic forms). The four harmonic forms are explicit:

- The constant 0-form v_0 (spanning H^0);
- The two independent harmonic 1-forms v_1^A, v_1^B (spanning H^1);
- The constant 2-form v_2 (spanning H^2).

These four forms are the four emergent spacetime directions (see HCSM-03 for the Lorentzian-signature derivation).

Corollary 6.3 (Emergent dimension). *The dimension of the emergent spacetime is $d = \dim \ker D = 4$. The Hodge complex selects four dimensions intrinsically.*

7 The D_4 decomposition of the $\lambda = 4$ sector

7.1 The D_4 action on the Hodge complex

The dihedral group D_4 of order 8 acts on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ by the point group of the square, generated by the rotation

$$r : (x_1, x_2) \mapsto (-x_2, x_1) \quad (25)$$

and the reflection

$$s : (x_1, x_2) \mapsto (x_1, -x_2), \quad (26)$$

subject to the presentation $r^4 = s^2 = (rs)^2 = e$.

The action on the plane-wave basis $v_k(x) = e^{i\pi k \cdot x/4}$ is by permutation of the momentum labels $k \in \{0, \dots, 7\}^2$. The action on Ω^1 is the vector-twisted version: the components (ω_1, ω_2) transform as

$$r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1), \quad s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2). \quad (27)$$

As a D_4 representation, $\Omega^1 = \Omega^0 \otimes E$, where E is the two-dimensional irreducible representation of D_4 .

7.2 The D_4 decomposition

Theorem 7.1 (D_4 decomposition of the $\lambda = 4$ sector). *As D_4 representations, the three form degrees at $\lambda = 4$ decompose as*

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (28)$$

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (29)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E. \quad (30)$$

The full 56-dimensional sector decomposes as

$$56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (31)$$

Proof. Direct computation of the traces of the permutation matrices R (for r) and S (for s) on the $\lambda = 4$ sector, followed by multiplicity computation via the D_4 character table. The tensor-product identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ follows from the vector-twisted action (27). All claims are verified at machine precision in the companion repository. $\square$

Remark 7.2 (Character of the 14-mode multiplet). The character of $\Omega^0|_{\lambda=4}$ is

$$\chi(e) = 14, \quad \chi(r^2) = 2, \quad \chi(r) = 0, \quad \chi(s) = 2, \quad \chi(rs) = 2. \quad (32)$$

The vanishing of $\chi(r)$ is a structural feature: the cyclic rotation r has no fixed content on the multiplet. This is the geometric reason D_4 acts effectively as $\mathbb{Z}_2 \times \mathbb{Z}_2$ on the 14 modes (see HCSM-04 for the orbit interpretation).

Remark 7.3 (The tensor-product structure). The identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ is a structural consequence of the Hodge construction: the 1-form components (e_1, e_2) transform in the vector representation E , so the Hodge complex extends the scalar D_4 representation by a single tensor factor of E in odd degree. This tensor structure is the geometric origin of the boson–fermion pairing.

8 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-02_verification`. It constructs the discrete Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the exterior derivatives d_0 and d_1 , the codifferentials d_0^* and d_1^* , the Hodge–Dirac operator D , and the D_4 representation matrices, and verifies:

1. The nilpotency $d_1 d_0 = 0$ at machine precision.
2. The Hodge–Dirac square identity $D^2|_{\Omega^k} = \Delta$ on each form degree $k = 0, 1, 2$.
3. The 14-fold lift: multiplicity 14 in Ω^0 , 28 in Ω^1 , 14 in Ω^2 at $\lambda = 4$.
4. The kernel dimension $\dim \ker D = 4$.
5. The $\mathbb{Z}_2$ grading: $D(-1)^F = -(-1)^F D$.
6. The supercharge algebra $\{Q, Q^\dagger\} = 2H$.
7. The D_4 decomposition of each form degree at $\lambda = 4$.
8. The tensor-product identity $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

9 Conclusion

This paper has established the Hodge complex of differential forms on the periodic 8×8 torus as the algebraic substrate of the Hodge Complex Standard Model. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree, in contrast to the spinorial Dirac operator on the same lattice. The complex carries a canonical $\mathbb{Z}_2$ grading — even forms bosonic, odd forms fermionic — and realizes an $\mathcal{N} = 1$ supersymmetric quantum mechanics with D as supercharge.

Three structural theorems anchor the paper. The 14-fold degeneracy at $\lambda = 4$ lifts to $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$ modes. The kernel of D has dimension $b_0 + b_1 + b_2 = 4$, giving the four emergent spacetime directions. The D_4 decomposition of the $\lambda = 4$ sector is $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$, with $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ as a tensor-product identity.

The Hodge complex is the substrate on which every downstream HCSM result is built. It is D_4 -symmetric, entropic (see HCSM-12 for its flat thermal output), and provides the 14-mode scalar multiplet at $\lambda = 4$ that the FDLM/WIN processor reads out.

References

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A Web Crawler Metadata

A.1 Dublin Core

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 DC.date.issued: 2026-09
 DC.type: Text / Preprint / White Paper
 DC.format: application/pdf; text/x-latex
 DC.language: en

DC.publisher: Entropia Research Institute
 DC.rights: All rights reserved
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 DC.identifier: HCSM-02-2026-09

A.2 Schema.org JSON-LD

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    complex of differential forms on  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ . The Hodge-Dirac
    operator  $D = d + d^*$  satisfies  $D^2 = \Delta$  exactly on each form degree,
    in contrast to the spinorial Dirac operator. The complex carries a
    canonical  $\mathbb{Z}_2$  grading (even forms bosonic, odd forms fermionic) and
    realizes an  $N=1$  supersymmetric quantum mechanics. The 14-fold
    degeneracy at  $\lambda=4$  lifts to  $14 \Omega^0 + 28 \Omega^1 + 14 \Omega^2$ 
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    = 56 modes. The kernel dimension is  $\dim \ker D = b_0 + b_1 + b_2 = 4.$ ",
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    "Z2 grading",
    "supercharge",
    "supersymmetric quantum mechanics",
    "8x8 torus",
    "kernel dimension",
    "D4 representation theory"
  ],
  "license": "All rights reserved",
  "identifier": "HCSM-02-2026-09",
  "isPartOf": {
    "@type": "Periodical",
    "name": "HCSM White Paper Series"
  }
}

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A.3 BibTeX for Automated Indexing

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@article{PreschuttiHCSM02,
  author = {Preschutti, Stanley},
  title = {{HCSM-02}: The Hodge Complex on  $\Lambda$ },
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, Hodge complex,
    differential forms, Hodge-Dirac operator,  $D^2 = \Delta$ ,
    Z2 grading, supercharge, supersymmetric quantum mechanics,
    8x8 torus, kernel dimension, D4 representation theory},
  abstract = {This paper establishes the Hodge complex of differential
    forms on the periodic 8x8 torus as the algebraic substrate
    of the Hodge Complex Standard Model. The Hodge-Dirac
    operator  $D = d + d^*$  satisfies  $D^2 = \Delta$  exactly on each
    form degree. The 14-fold degeneracy at  $\lambda=4$  lifts to
     $14 \Omega^0 + 28 \Omega^1 + 14 \Omega^2 = 56$  modes. The kernel
    dimension is  $\dim \ker D = b_0 + b_1 + b_2 = 4.$ }
}

```